Hussain Singapuri

AI ENGINEER (AGENTIC & GENERATIVE AI)

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PROFESSIONAL SUMMARY

AI Engineer (Agentic & Generative AI) with 3 years of professional full-stack experience at a US-based medical-device company, including AI agents built for internal use. Built agents and RAG end to end: site-to-agent (hybrid BM25 + vector retrieval on pgvector, eval harness) and multi-agent-research-crew (tool-calling agents on a custom DAG orchestrator). Strong Python and FastAPI, Claude and Gemini APIs, webhooks and CRM integrations, and Claude Code used daily to ship faster.

CORE COMPETENCIES

Agentic AI | RAG | LLM APIs | Python | FastAPI | pgvector | Function calling | Webhooks and CRM integrations

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built AI agents for internal use
- Built several apps, including a CRM
- Built and maintained full-stack web applications for the company
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

site-to-agent

RAG agent end to end: crawler, hybrid BM25 + vector retrieval with RRF on pgvector, cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

multi-agent-research-crew

Planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking for hallucination control.

Python, asyncio, Claude

ai-lead-receptionist

Chat and SMS agent that qualifies leads and books jobs, with missed-call text-back and an owner dashboard; integrates telephony (Twilio SMS) with a database.

Next.js, TypeScript, Twilio, Claude, Supabase

nanoserve

LLM inference engine built from scratch with paged KV cache, continuous batching, speculative decoding and an OpenAI-compatible streaming API; 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

PageForge

AI drafts only from approved facts; a deterministic validator flags unsupported numbers and a reviewer signs off claim by claim. Events flow to PostHog and HubSpot.

Next.js 16, React 19, Sanity, Supabase, Gemini, PostHog, HubSpot

TripSync

AI business-travel planner that returns three AI-generated packages via Gemini structured outputs and refines them via chat.

Flutter, Supabase (RLS, Edge Functions), Gemini structured outputs

More projects

lead-intel-engine, contractor-ai-ops-dashboard, Personal portfolio with full-stack backend, Orbital Surplus, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

CERTIFICATIONS

Claude Certified Developer - Foundations (CCDV-F)	Anthropic
Claude Certified Orchestrator - Foundations (CCAO-F)	Anthropic
Claude Code in Action	Anthropic
Claude Code 101	Anthropic
Generative AI Leader	Google Cloud

SKILLS

AI / LLM: AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude API, Gemini API, Claude Code and AI coding agents

Languages: Python, SQL, TypeScript, JavaScript, Dart, HTML, CSS

Back end: FastAPI, asyncio, REST APIs, webhooks (HMAC signing), Node.js, Stripe Checkout and refunds, auth (scrypt, sessions, throttling), Resend

Data: pgvector, PostgreSQL, SQLite, Neon, Supabase (RLS, Edge Functions), PGlite, Sanity CMS

Automation & integrations: workflow automation, CRM integrations (HubSpot), Twilio SMS, PostHog, web crawling / scraping

DevOps: Docker, GitHub Actions, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Front end: Next.js, React 19, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO